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Credit Cycles, Expectations, and Corporate Investment

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Expectations data and construction of biased-expectations measures	3
2.2	Firm-level, credit-market, and macroeconomic data	4
2.3	Outcomes and controls	4
3	Models and empirical specifications (all core equations + interpretation)	5
3.1	Forecast revisions and forecast errors	5
3.2	Predictable component of forecast errors	5
3.3	Aggregating firm-level bias to the macro level	6
3.4	Predicting credit market sentiment	6
3.5	Local projections for investment and debt issuance	6
3.6	Cross-sectional asymmetry by optimistic and pessimistic firms	7
3.7	Baseline Q-theory problem	7
3.8	First-order conditions and the investment equation	8
3.9	Financial frictions	8
3.10	Diagnostic expectations and endogenous reversals	8
3.11	Risky debt and credit-supply effects	9
4	Identification and credibility	10
4.1	What the paper uses in place of a classical instrument	10
4.2	Why debt-weighting is crucial	10
4.3	Validation exercises	10
4.4	Separating credit supply from credit demand	11
4.5	Inference and controls	11
4.6	Main limitation	11
5	Tables and figures	11

5.1	Table 1: Predicting forecast errors with past forecast revisions	11
5.2	Figure 1: CMS and predicted forecast error	12
5.3	Table 2: Predicting credit market sentiment with predictable forecast errors	12
5.4	Figure 2: CMS and aggregate corporate credit-market measures	14
5.5	Table 3: Validation of firmPFE and macroPFE	14
5.6	Table 4: Predictable forecast errors drive cycles in corporate investment	15
5.7	Table 5: Predictable forecast errors drive cycles in debt issuance	15
5.8	Table 6: Investment cycles conditioning on financial constraints	16
5.9	Table 7: Debt-issuance cycles conditioning on financial constraints	16
5.10	Table 8: Physical-investment cycles conditioning on firm-level forecast-error terciles .	17
5.11	Table 9: Debt-issuance cycles conditioning on firm-level forecast-error terciles	17
5.12	Figure 3: Impulse response functions from Q-theory	17
6	Short summary	18
7	Conclusion	18
7.1	Core conclusion (what the paper establishes)	18
7.2	Economic intuition	19
7.3	Takeaway	19

1 Big picture

- **Question.** Do irrational swings in expectations about firm fundamentals drive credit booms and busts, and do those credit cycles in turn drive corporate borrowing and real investment?
- **Setting.** U.S. nonfinancial publicly traded firms. Expectations data come from IBES long-term earnings-growth forecasts; firm outcomes come from Compustat; the main annual sample for firm-level tests runs from 1990 to 2017.
- **Core empirical object.** The paper builds an aggregate expectations index, macroPFE_t , by taking the predictable component of firm-level long-term forecast errors and weighting it by contemporaneous firm-level debt issuance.
- **Main aggregate result.** The new index closely tracks credit market sentiment. Its correlation with the Greenwood–Hanson issuer-quality sentiment measure is 0.75, and it predicts a credit boom in the near term and a reversal three to five years later.
- **Main firm-level result.** A one-standard-deviation increase in macroPFE_t predicts a 4.6% rise in total investment in year $t + 1$ and a 2.7% and 2.5% decline in years $t + 4$ and $t + 5$. It also predicts a much larger boom-bust cycle in long-term debt issuance: +24.0% in year $t + 1$, +13.4% in year $t + 2$, then −12.3% and −10.3% in years $t + 4$ and $t + 5$.
- **Supply versus demand.** The boom-bust responses of investment and debt issuance are almost identical for financially constrained and unconstrained firms. That pattern is inconsistent with a pure supply-only story and instead supports Minsky’s idea that both investors and managers become too optimistic.
- **Cross-sectional mechanism.** When aggregate optimism is high, firms with the most optimistic predictable firm-level expectations exhibit the strongest issuance and investment booms and the sharpest later reversals. When aggregate optimism is low, the mirror image holds for the most pessimistic firms.
- **Theory.** The paper argues that standard rational-expectations financial-frictions models can explain short-run amplification but struggle to explain predictable reversals and the lack of heterogeneity by financial constraints. A dynamic Q-theory model with diagnostic expectations by both managers and credit investors fits the qualitative facts better.
- **Why the paper matters.** The paper turns the Minsky hypothesis into a measurable empirical program. It connects analyst forecast revisions, credit-market sentiment, debt issuance, and investment in one unified framework, and it gives a policy-relevant answer: credit booms look like periods of biased expectations rather than merely rational responses to fundamentals.

2 Data and measurement (key objects)

2.1 Expectations data and construction of biased-expectations measures

- **Analyst expectations.** The paper uses analyst-level forecasts from the IBES Unadjusted Detail History File. The key expectation is long-term earnings-growth, LTG, defined by IBES as the expected annual increase in operating earnings over the next full business cycle, approximately 3 to 5 years.

- **Frequency conversion.** Monthly consensus forecasts are aggregated to the annual level using firm-year averages of median monthly forecasts.
- **Realizations.** Realized earnings per share come from the IBES Unadjusted Detail Actuals file, with stock-split adjustments from CRSP daily data.
- **Time span.** LTG data run from 1983 through 2017, which is long enough to estimate multi-year forecast errors and rolling predictive relations.
- **Main firm-level expectation object.** The paper extracts the predictable part of forecast errors from a regression of future forecast errors on the previous two forecast revisions. The fitted value is the firm-level measure of biased expectations, $\text{firmPFE}_{i,t}$.
- **Why this variable matters.** Under rational expectations, forecast errors should not be forecastable from lagged revisions. Predictability therefore indicates systematic over- or underreaction to news.

2.2 Firm-level, credit-market, and macroeconomic data

- **Firm sample.** The annual firm panel uses Compustat from 1990 to 2017 and excludes financial firms, utilities, non-U.S. firms, and firm-years with negative assets, sales, or book equity.
- **Sample size.** The paper reports more than 70,000 firm-year observations and 8,286 unique firms, with a median of 2,543 firms per year. The main local-projection specifications use about 56,000 firm-years because they include lagged dependent variables.
- **Credit market sentiment.** The baseline sentiment measure is the Greenwood–Hanson issuer-quality index, ISS, defined as the difference between the average default probability of high net-debt issuers and that of low net-debt issuers.
- **Default probabilities.** Firm-level default probabilities follow Bharath and Shumway (2008), which lets the paper extend the credit-sentiment measure back before the broad availability of ratings data.
- **Alternative credit-sentiment variables.** The paper also studies the high-yield-share index (HYS), the default spread, Gilchrist–Zakrajsek credit spreads and excess bond premia, and Schularick–Taylor bank-loan measures.
- **Macro controls.** The main aggregate controls are the Leading Economic Index, the Jurado–Ludvigson–Ng macro uncertainty index, the Baker–Wurgler equity sentiment index, the default spread, the risk-free rate, and the Baker–Bloom–Davis policy-uncertainty index.
- **Validation data.** To validate the expectations index, the paper also uses firm-level excess bond premia from Gilchrist and Zakrajsek (2012) and Blue Chip Financial Forecasts for future BAA yields.

2.3 Outcomes and controls

- **Investment measures.** Following Peters and Taylor (2017), total capital equals physical capital plus intangible capital. Physical capital is gross PPE; intangible capital is goodwill plus capitalized R&D plus capitalized SG&A.

- **Dependent variables on the real side.** The paper studies total investment, physical investment, and intangible investment, each scaled by lagged total capital.
- **Debt issuance measures.** The main financing variable is the change in long-term debt ($dltt + dlc$) scaled by lagged assets. The paper also splits debt into portions due in more than one year and in less than one year.
- **Firm controls.** Tobin’s q , cash flow to assets, log assets, cash to assets, book leverage, sales growth, ROA, and default probability are included to absorb time-varying investment opportunities and balance-sheet strength.
- **Financial-constraint measures.** To separate supply from demand mechanisms, the paper uses the Hadlock–Pierce, Whited–Wu, and Kaplan–Zingales measures of financial constraints.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Forecast revisions and forecast errors

The firm-level forecast revision from year $t - 1$ to year t is

$$FR_{i,t-1 \rightarrow t} = E_t[LTG] - E_{t-1}[LTG]. \quad (1)$$

The corresponding multi-year forecast error is

$$FE_{i,t \rightarrow t+k} = \left(\frac{EPS_{t+k}}{EPS_t} \right)^{1/k} - E_t[LTG], \quad k = 1, \dots, 5. \quad (2)$$

- **Interpretation.** FR measures how much analysts revise long-run earnings-growth expectations between two adjacent years. FE compares the annualized realized EPS growth over horizon k with the forecast made at t .
- **Expected sign under rational expectations.** If expectations are rational, lagged forecast revisions should not predict future forecast errors.
- **Economic idea.** Positive revisions followed by negative forecast errors are evidence of overextrapolation: analysts become too optimistic after good news.

3.2 Predictable component of forecast errors

The paper estimates rolling predictive regressions of the form

$$FE_{i,t \rightarrow t+k} = \alpha_{t+k} + \beta_{1,t} FR_{i,t-2 \rightarrow t-1} + \beta_{2,t} FR_{i,t-1 \rightarrow t} + \epsilon_{i,t+k}, \quad k = 1, \dots, 5. \quad (3)$$

The fitted value at the four-year horizon defines the firm-level biased-expectations measure:

$$\text{firmPFE}_{i,t} = \hat{\alpha}_t + \hat{\beta}_{1,t} FR_{i,t-2 \rightarrow t-1} + \hat{\beta}_{2,t} FR_{i,t-1 \rightarrow t}. \quad (4)$$

- **Interpretation.** firmPFE is the portion of future forecast error that was already predictable using the last two revisions. The paper interprets it as firm-level overextrapolation of shocks to fundamentals.
- **Rolling estimation.** For each year t , the regression uses only data available up to t , so the fitted value is genuinely real-time and does not use future information.
- **Key empirical sign.** Table 1 shows negative β 's, especially at horizons two through five years, meaning optimistic revisions are followed by disappointments.

3.3 Aggregating firm-level bias to the macro level

The aggregate expectations index is

$$\text{macroPFE}_t = \sum_{i=1}^N w_{i,t} (-\text{firmPFE}_{i,t}), \quad (5)$$

with weights given by contemporaneous debt issuance,

$$w_{i,t} = \frac{\Delta \text{Debt}_{i,t-1 \rightarrow t}}{\text{Assets}_{i,t-1}}. \quad (6)$$

- **Interpretation.** The paper does not simply average biased expectations across firms. It gives more weight to firms that are actually accessing credit markets in that year.
- **Why the minus sign?** Forecast errors are more negative in optimistic booms, so the minus sign makes high optimism correspond to high macroPFE.
- **Minsky logic.** Credit cycles matter when optimistic firms borrow. Debt-weighting is therefore not a cosmetic choice; it is central to the mechanism.

3.4 Predicting credit market sentiment

The aggregate predictive regression is

$$\text{CMS}_{t+k} = \alpha + \beta \text{macroPFE}_t + \epsilon_t, \quad k = 0, \dots, 5. \quad (7)$$

- **Interpretation.** A positive coefficient at short horizons means biased expectations coincide with a credit boom. A negative coefficient at longer horizons means the boom later reverses.
- **Key empirical pattern.** The paper finds +0.750 contemporaneously, +0.484 at $t+1$, then -0.367, -0.558, and -0.434 at $t+3$, $t+4$, and $t+5$.
- **Comparison object.** The equally weighted average of firmPFE has much weaker explanatory power, showing that credit-market access is crucial for turning biased beliefs into sentiment.

3.5 Local projections for investment and debt issuance

The main firm-level empirical specification is a panel local projection:

$$Y_{i,t+k} = \alpha_i + \beta_k \text{macroPFE}_t + \gamma' F_{i,t} + \delta' M_t + \rho_0 Y_{i,t} + \rho_1 Y_{i,t-1} + \varepsilon_{i,t+k}, \quad k = 1, \dots, 5. \quad (8)$$

- **Dependent variables.** $Y_{i,t+k}$ is total investment, physical investment, intangible investment, or one of several debt-issuance measures.
- **Controls.** $F_{i,t}$ are firm controls and M_t are macro controls. Firm fixed effects absorb time-invariant heterogeneity.
- **Interpretation of β_k .** Each β_k is the impulse response of the firm outcome k years after a one-standard-deviation increase in aggregate biased expectations.
- **Inference.** Because macroPFE $_t$ is common to all firms in year t , the paper bootstraps standard errors following Abadie et al. (2023) to account for both cross-sectional and serial dependence.

3.6 Cross-sectional asymmetry by optimistic and pessimistic firms

To test whether firms with stronger own biased expectations respond more over the cycle, the paper estimates

$$Y_{i,t+k} = \sum_{j=1}^3 \beta_j^H \text{HighMacroPFE}_t \cdot \mathbf{1}\{T_{i,t} = j\} + \sum_{j=1}^3 \beta_j^L \text{LowMacroPFE}_t \cdot \mathbf{1}\{T_{i,t} = j\} + \delta X_{i,t} + \epsilon_t, \quad (9)$$

where $T_{i,t}$ indexes the tercile of firmPFE $_{i,t}$.

- **Interpretation.** This design compares high- and low-firmPFE firms separately in optimism states and pessimism states.
- **Prediction.** If Minsky-style biased expectations matter, high-firmPFE firms should show stronger booms and stronger later reversals when aggregate optimism is high, while low-firmPFE firms should show the mirror image when aggregate optimism is low.

3.7 Baseline Q-theory problem

The theoretical firm chooses labor, investment, and next-period capital to maximize expected discounted earnings:

$$\max_{\{I_s, L_s, K_{s+1}\}_{s=t}^{\infty}} E_t \left[\sum_{s \geq t} \beta^{s-t} \left(A_s K_s^\alpha L_s^{1-\alpha} - \omega_s L_s - I_s - C(I_s, K_s) K_s \right) \right], \quad (10)$$

subject to

$$K_{s+1} = (1 - \delta)K_s + I_s, \quad (11)$$

and quadratic adjustment costs

$$C(I_s, K_s) = \frac{\chi}{2} \left(\frac{I_s}{K_s} - \delta \right)^2. \quad (12)$$

- **Interpretation.** This is a standard neoclassical investment problem. In the baseline model, fundamentals move investment through expected future profits.
- **Where expectations enter.** The expectations operator $E_t(\cdot)$ is allowed to reflect biased beliefs, provided it preserves the law of iterated expectations.

3.8 First-order conditions and the investment equation

The core first-order conditions are

$$(1 - \alpha)A_t K_t^\alpha L_t^{1-\alpha} = \omega_t, \quad (13)$$

$$q_t - 1 - \chi \left(\frac{I_t}{K_t} - \delta \right) = 0, \quad (14)$$

and

$$q_t = \beta E_t \left[\alpha A_{t+1} K_{t+1}^{\alpha-1} L_{t+1}^{1-\alpha} + \chi \frac{I_{t+1}}{K_{t+1}} \left(\frac{I_{t+1}}{K_{t+1}} - \delta \right) - \frac{\chi}{2} \left(\frac{I_{t+1}}{K_{t+1}} - \delta \right)^2 + q_{t+1}(1 - \delta) \right]. \quad (15)$$

After forward iteration, the standard Q-theory investment condition is

$$\frac{I_t}{K_t} = \delta - \frac{1}{\chi} + \frac{\beta}{\chi} E_t \left[\frac{\sum_{s \geq t+1} \beta^{s-(t+1)} \Pi_s}{K_{t+1}} \right]. \quad (16)$$

- **Interpretation.** Investment is high when expected discounted future profits are high relative to installed capital.
- **Link to the empirics.** This investment equation is the theoretical counterpart to the local-projection regressions: expectation shocks affect investment because they change expected future profits.

3.9 Financial frictions

The paper then adds risky borrowing and a smooth approximation to a collateral constraint. Borrowing carries a tax advantage τB_t but is costly through

$$C^D(B_s, K_s) = \phi_0 \exp \left[-\phi_1 \left(\frac{\eta K_s}{B_s} - 1 \right) \right]. \quad (17)$$

- **Interpretation.** The formulation convexifies the occasionally nonbinding collateral constraint $B_t \leq \eta K_t$.
- **Economic meaning.** Higher η means more pledgeable capital. Large ϕ_1 sharply penalizes debt above the collateral value.
- **Key limitation.** By itself this framework amplifies shocks but tends to imply that more constrained firms should react more strongly, which is not what the data show.

3.10 Diagnostic expectations and endogenous reversals

The paper approximates the investment rule by

$$i_t = b_0 + b_1 E_t(\pi_{t+1}), \quad (18)$$

where lowercase variables are scaled by capital. If profits follow

$$\pi_{t+1} = \rho\pi_t + \epsilon_{t+1}, \quad (19)$$

then under rational expectations,

$$i_t = b_0(1 - \rho) + \rho i_{t-1} + b_1 \rho \epsilon_t. \quad (20)$$

Under diagnostic expectations,

$$E_t^\theta(\pi_{t+1}) = E_t(\pi_{t+1}) + \theta[E_t(\pi_{t+1}) - E_{t-1}(\pi_{t+1})], \quad (21)$$

which implies

$$i_t = b_0(1 - \rho) + \rho i_{t-1} + b_1 \rho(1 + \theta)\epsilon_t - b_1 \theta \rho^2 \epsilon_{t-1}. \quad (22)$$

The paper further shows

$$E_t[i_{t+1} - E_t^\theta(i_{t+1})] = -b_1 \theta \rho^2 \epsilon_t, \quad (23)$$

and more generally

$$E_t[E_{t+k}^\theta(i_{t+T}) - E_t^\theta(i_{t+T})] = -b_1 \theta \rho^{T+1} \epsilon_t. \quad (24)$$

- **Interpretation.** Diagnostic expectations add a moving-average component to the investment response. Positive news produces a large short-run boom and then a predictable reversal once beliefs are revised back.
- **Empirical payoff.** This is exactly the shape the paper documents in investment, debt issuance, and credit sentiment.

3.11 Risky debt and credit-supply effects

To get both demand and supply effects, the paper lets credit investors also form diagnostic expectations. With risky debt,

$$E[g] = r = \pi_s r_B + \pi_f \cdot 0, \quad (25)$$

so the promised risky debt rate is

$$r_B = \frac{E[g]}{\pi_s} = \frac{r}{\pi_s}. \quad (26)$$

If default occurs when productivity falls below a threshold A^* , then under rational expectations

$$\pi_{s,t+1} = 1 - \Phi \left(\frac{\log A^* - \rho \log A_t}{\sigma} \right), \quad (27)$$

whereas under diagnostic expectations

$$\pi_{s,t+1} = 1 - \Phi \left(\frac{\log A^* - \rho \log A_t - \theta \rho \epsilon_t}{\sigma} \right). \quad (28)$$

- **Interpretation.** If only managers are biased, good news raises demand for financing but should raise rates and reduce borrowing. To match the data’s falling credit spreads and rising debt issuance in booms, creditors must also become too optimistic about default risk.
- **Model implication.** Joint diagnostic expectations of managers and creditors create the largest boom-bust responses in both I/K and B/K .

4 Identification and credibility

4.1 What the paper uses in place of a classical instrument

- **No external shock design.** This is not an IV or natural-experiment paper. Its empirical leverage comes from measuring the predictable component of expectation errors, validating that measure in credit markets, and then studying its dynamic consequences.
- **Core identification idea.** Under rational expectations, lagged forecast revisions should not predict future forecast errors. The fact that they do predict forecast errors — and with the overextrapolation sign — is the first identifying step.
- **Real-time implementation.** Rolling estimation using only information available up to year t avoids look-ahead bias in constructing firmPFE and macroPFE.

4.2 Why debt-weighting is crucial

- **Minsky mechanism.** Minsky’s theory is about optimistic firms borrowing too much and then cutting back later. Debt-weighting therefore targets the subset of firms whose biased expectations are most relevant for credit conditions.
- **Empirical confirmation.** The equal-weighted average firmPFE has only a 0.29 correlation with CMS, versus 0.75 for the debt-weighted index.
- **Interpretation.** The paper’s aggregate index is not “general optimism”; it is optimism among the firms actually financing in credit markets.

4.3 Validation exercises

- **Firm-level credit pricing.** firmPFE is positively related to firm-level excess bond premia, with coefficients of about 0.082, 0.059, and 0.058 depending on aggregation level.
- **Direct forecast evidence.** macroPFE predicts 12-, 15-, and 18-month-ahead BAA-yield forecast errors with coefficients around 0.67 to 0.69.

- **Alternative sentiment measures.** The same boom-bust pattern appears for HYS, the default spread, Gilchrist–Zakrajsek spreads, and bank-loan measures.

4.4 Separating credit supply from credit demand

- **Key thought experiment.** If only supply shifted and managers were rational, unconstrained firms should mostly arbitrage cheap debt by recapitalizing rather than investing more. If only demand shifted and investors were rational, credit spreads should rise. The data do not fit either pure case.
- **Empirical test.** The paper compares constrained and unconstrained firms using three financial-constraint measures. The interaction of macroPFE with financial constraints is generally small and insignificant for both investment and debt issuance.
- **Conclusion.** The similarity of responses across constrained and unconstrained firms is the paper’s most direct evidence that both demand-side and supply-side optimism are present.

4.5 Inference and controls

- **Firm-side controls.** Rich balance-sheet and profitability controls reduce the scope for the results to be driven by time-varying investment opportunities, liquidity, or default risk.
- **Macro controls.** Leading indicators, uncertainty, equity sentiment, rates, and policy uncertainty are all controlled for, so the paper is not just repackaging a standard business-cycle predictor.
- **Error structure.** Because macroPFE varies only over time, the paper uses bootstrapped standard errors tailored to common shocks in the panel local-projection setting.

4.6 Main limitation

- **Interpretation is disciplined but not fully causal in the experimental sense.** The paper offers strong, theory-guided predictive evidence and multiple validation checks, but it does not isolate an exogenous shock to beliefs with a separate instrument.
- **Why the evidence is still persuasive.** The sign patterns, debt-weighting result, validation against bond-market variables, similar constrained/unconstrained responses, and the cross-sectional symmetry all line up with one coherent mechanism.

5 Tables and figures

5.1 Table 1: Predicting forecast errors with past forecast revisions

Read:

- Forecast revisions are strongly negatively related to future forecast errors at horizons two through five years. For example, the coefficient on $FR_{t-1 \rightarrow t}$ is about -0.567 at the two-year horizon and about -0.644 at the four-year horizon.
- The sign is the overextrapolation sign: optimistic revisions are followed by disappointments.

- The table is the empirical foundation for everything that follows, because it justifies treating the fitted component of forecast errors as biased expectations.

Table 1
Predicting forecast errors with past forecast revisions

	$FE_{t \rightarrow t+1}$	$FE_{t \rightarrow t+2}$	$FE_{t \rightarrow t+3}$	$FE_{t \rightarrow t+4}$	$FE_{t \rightarrow t+5}$
$FR_{t-1 \rightarrow t}$	-0.191 (-1.32)	-0.567*** (-5.22)	-0.623*** (-8.78)	-0.644*** (-11.01)	-0.638*** (-14.74)
$FR_{t-2 \rightarrow t-1}$	-0.465*** (-3.89)	-0.528*** (-4.89)	-0.500*** (-6.92)	-0.475*** (-8.28)	-0.480*** (-13.18)
Observations	37,856	32,345	29,563	26,960	24,697
R^2	.001	.014	.029	.045	.060

This table presents regressions of $FE_{t,t \rightarrow t+k}$ on two forecast revisions. Each column corresponds to a different $k=1, \dots, 5$. Forecast errors and revisions are calculated using forecasts of long-term EPS growth (LTG) from IBES. See Section 2 for details on their construction. Standard errors are corrected for autocorrelation using the Newey and West (1987) procedure. t -statistics are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

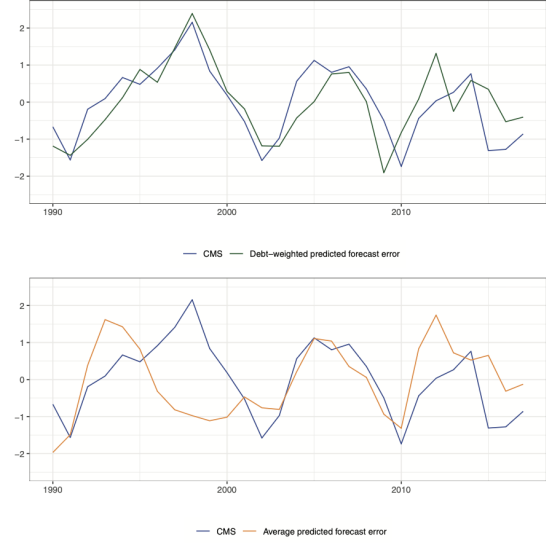


Figure 1
CMS and predicted forecast error
This figure plots the credit market sentiment (CMS) index of Greenwood and Hanson (2013) in blue alongside the debt-weighted firm-level predicted forecast error (*macroPFE*, top panel, green) and average firm-level predicted forecast error (*firmPFE*, bottom panel, orange). All variables are standardized to have a standard deviation of 1 and mean of 0.

5.2 Figure 1: CMS and predicted forecast error

Read:

- The top panel shows that the debt-weighted macroPFE series and CMS move together very closely over time.
- The bottom panel shows that the equally weighted average firmPFE is much noisier and tracks CMS far less well.
- Visually and quantitatively, the figure makes the case that credit-market optimism is specifically tied to optimistic firms issuing debt, not to average analyst optimism in the economy.

5.3 Table 2: Predicting credit market sentiment with predictable forecast errors

Read:

- **Panel A.** The debt-weighted index predicts a classic boom-bust cycle in CMS: +0.750 contemporaneously, +0.484 at $t+1$, then -0.367, -0.558, and -0.434 in years $t+3$ to $t+5$.
- **Panel B.** The equal-weighted index is much weaker; it has little explanatory power for later reversals.
- **Takeaway.** The paper's new expectations measure is not only correlated with credit sentiment; it predicts its reversal several years ahead.

Table 2

Predicting CMS with predictable firm-level forecast error

	CMS_t	CMS_{t+1}	CMS_{t+2}	CMS_{t+3}	CMS_{t+4}	CMS_{t+5}
A. macroPFE is firmPFE weighted by total debt issuance (change in long-term debt)						
macroPFE	0.750*** (7.942)	0.484*** (4.583)	0.035 (0.256)	-0.367** (-2.471)	-0.558*** (-7.507)	-0.434*** (-3.737)
Constant	-0.000 (-0.000)	0.017 (0.089)	0.085 (0.363)	0.106 (0.415)	0.097 (0.362)	0.077 (0.306)
Observations	28	27	26	25	24	23
R ²	.563	.237	.001	.150	.341	.208
B. firmPFE is a simple average of firmPFE						
firmPFE	0.287 (1.338)	0.299* (1.907)	0.166* (1.803)	0.063 (0.237)	-0.098 (-0.256)	-0.067 (-0.237)
Constant	0.000 (0.000)	0.024 (0.095)	0.083 (0.342)	0.098 (0.387)	0.094 (0.356)	0.069 (0.262)
Observations	28	27	26	25	24	23
R ²	.082	.091	.031	.004	.011	.005

This table presents regression of CMS_{t+k} on an aggregation of $firmPFE$, which is calculated differently in each panel. In panel A, $macroPFE$ is calculated by weighting $firmPFE$ by change in total debt in the last year ($dltr_t + dlcr_t - dltr_{t-1} - dlcr_{t-1}$, scaled by total assets at $t-1$). In panel B, $\bar{firmPFE}$ is a simple average annually of $firmPFE$. The predicted forecast errors are fitted values from panel regressions of forecast errors on forecast revisions. See Section 2 for details. Each column corresponds to a different $k=0, \dots, 5$. All variables are standardized so the economic magnitude of the coefficients is in terms of number of standard deviations from the mean. Standard errors are corrected for autocorrelation for up to six lags using the Newey and West (1987) procedure. t -statistics are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

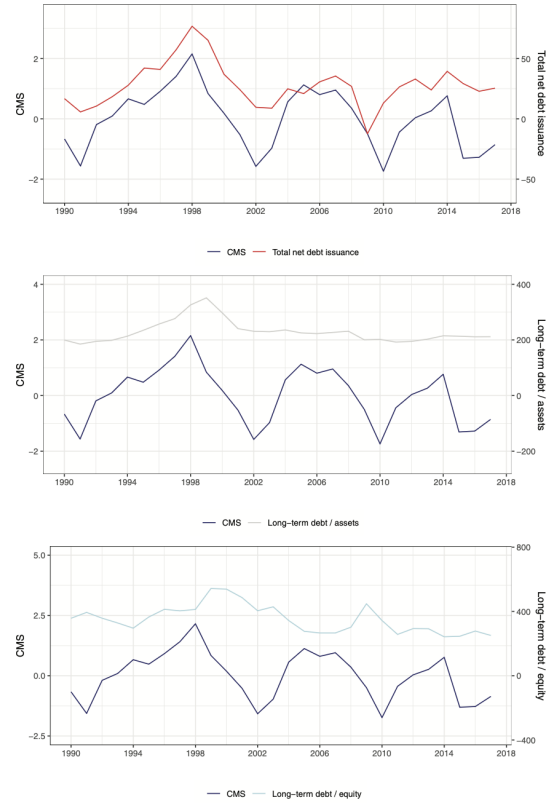


Figure 2

CMS and aggregate corporate credit market measures

This figure plots the credit market sentiment (CMS) index of Greenwood and Hanson (2013) in blue alongside three measures of corporate debt activity aggregated annually across the economy: (a) total net debt issuance ($dltr_t + dlcr_t - dltr_{t-1} - dlcr_{t-1}$) scaled by total assets at $t-1$ in panel A, (b) total long-term debt ($dltr_t + dlcr_t$) scaled by total assets at $t-1$ in panel B, and (c) total long-term debt ($dltr_t + dlcr_t$) scaled by the market value of equity at $t-1$ in panel C. All variables are standardized to have a standard deviation of 1 and mean of 0.

5.4 Figure 2: CMS and aggregate corporate credit-market measures

Read:

- Total net debt issuance rises and falls with CMS, which supports the idea that credit booms are issuance booms.
- Long-term debt scaled by assets also rises with CMS, but long-term debt scaled by equity is flatter and sometimes falls while sentiment is high.
- The key interpretation is that CMS is not simply equity-market sentiment in disguise. When sentiment is high, debt issuance rises but market equity also rises, so market leverage does not mechanically explode.

5.5 Table 3: Validation of firmPFE and macroPFE

Read:

- **Panel A.** firmPFE is positively related to firm-level excess bond premia at the bond-month, firm-month, and firm-year levels. That means the expectation-bias measure lines up with credit-market mispricing.
- **Panel B.** macroPFE strongly predicts future BAA-yield forecast errors, with coefficients between 0.668 and 0.691.
- **Takeaway.** These validation tests tie the expectations measure directly to credit-market expectations and pricing, not just to accounting outcomes.

Table 3
Validation of firmPFE and macroPFE

A. Regressions of excess bond premium on predictable forecast errors

	Bond-month-level	Firm-month-level	Firm-year-level
<i>firmPFE</i>	0.082*** (5.41)	0.059*** (4.57)	0.058*** (2.91)
Firm fixed effects	Yes	Yes	Yes
Observations	335,391	86,569	7,967
R ²	.158	.239	.317

B. Predicting BAA forecast errors with macroPFE

	<i>Error_{t→t+12}</i>	<i>Error_{t→t+15}</i>	<i>Error_{t→t+18}</i>
<i>macroPFE</i>	0.686*** (5.65)	0.691*** (4.91)	0.668*** (3.87)
Observations	18	18	18
R ²	.471	.477	.446

Panel A reports coefficient estimates from regressing excess bond premia (EBP) on *firmPFE*. The dependent variable EBP is measured at the bond-month level following Gilchrist and Zakrajsek (2012), and *firmPFE* is measured at the firm-year level. In the first column of panel A, we run regressions at the bond-month level, where annual *firmPFE* is simply duplicated for all bonds of the firm, in all months of the corresponding year. In the second column of panel A, we aggregate the dependent variable at the month level, using a weighted average of the EBP values of the firm's bonds that month. The weights are given by the bonds' amount outstanding that month. We then run firm-month-level regressions with this weighted average EBP as the dependent variable and annual *firmPFE* values duplicated for all months of the corresponding year. Finally, in the third column of panel A, we run regressions at the firm-year level, where the dependent variable is a simple average of the monthly EBP averages used in the regressions with results reported in panel A, column 2. All regressions in panel A include firm fixed effects and cluster standard errors by firm and time. In panel B, we present coefficient estimates from regressions of 12-, 15-, and 18-months-ahead forecast errors of the BAA yield on *macroPFE*. Forecast errors are calculated each month *t* as the realized future BAA yield (12, 15, and 18 months into the future) and the consensus (median) forecast for it as of time *t*. Monthly forecast errors are averaged across each calendar year to match the frequency of our *macroPFE* index. Standard errors are corrected for autocorrelation using the Newey and West (1987) procedure. *t*-statistics are reported in parentheses. **p* < 0.1; ***p* < 0.05; ****p* < 0.01.

Table 4
Predictable firm-level forecast error drives cycles in corporate investment

	Year 1	Year 2	Year 3	Year 4	Year 5
A. Dependent variable: Total investment					
<i>macroPFE</i>	0.046*** (5.357)	0.044*** (5.349)	0.011 (1.259)	-0.027*** (-3.092)	-0.025*** (-3.010)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	57,597	52,648	47,150	42,139	37,643
R ²	.379	.314	.293	.288	.281
B. Dependent variable: Physical investment					
<i>macroPFE</i>	0.036*** (4.003)	0.012 (1.398)	-0.030*** (-3.547)	-0.079*** (-9.472)	-0.073*** (-8.294)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	57,597	52,658	47,712	42,664	38,160
R ²	.397	.351	.346	.335	.337
C. Dependent variable: Intangible investment					
<i>macroPFE</i>	0.047*** (4.447)	0.063*** (6.069)	0.036*** (3.380)	0.009 (0.825)	0.010 (0.828)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	57,709	52,747	47,245	42,223	37,721
R ²	.366	.309	.285	.282	.275

This table presents coefficient estimates from regressing total investment (panel A), investment in physical capital (panel B) and investment in intangible capital (panel C) up to 5 years in the future on aggregated predictable firm-level forecast error (*macroPFE*) and controls. Predictable firm-level forecast error (see Section 2 for details) is weighted by change in total debt (as in defined in panel A of Table 2). Column headings Year "k" (k = 1...5) mean the dependent variable is measured at time *t+k*, while all independent variables are measured at time *t*. The investment and Tobin's *q* variables are measured as in Peters and Taylor (2017). Specifically, total capital is gross PPE (i.e., physical capital) plus the sum of goodwill, capitalized R&D, and capitalized SG&A (i.e., intangible capital). Total investment is the percentage change in total capital, investment in physical capital is the change in physical capital divided by lagged total capital, and investment in intangible capital is the change in intangible capital divided by lagged total capital. Tobin's *q* is the market value of equity plus book value of debt divided by total capital. All specifications include firm fixed effects, firm-level controls (Tobin's *q*, Cash flow to assets, log total assets, Cash to assets, Book leverage, Sales growth, ROA, default probability) and macro-level controls (the Leading Economic Index from the Conference Board, the Jurado, Ludvigson, and Ng [2015] index of macro uncertainty, the Baker and Wurgler [2006] sentiment index, the default spread, the risk-free rate, and the policy uncertainty index of Baker, Bloom, and Davis [2016]), as well as the current and lagged level of the dependent variable. Standard errors are bootstrapped according to Abadie et al. (2023). *t*-statistics are reported in parentheses. **p* < 0.1; ***p* < 0.05; ****p* < 0.01.

5.6 Table 4: Predictable forecast errors drive cycles in corporate investment

Read:

- **Panel A.** Total investment rises 4.6% in year $t + 1$ and 4.4% in year $t + 2$, then falls 2.7% and 2.5% in years $t + 4$ and $t + 5$.
- **Panel B.** The long-run reversal is mainly a physical-capital story: physical investment falls 7.9% and 7.3% in years $t + 4$ and $t + 5$.
- **Panel C.** Intangible investment rises more persistently on the boom side and mostly fades rather than sharply reverses.
- **Takeaway.** Optimistic credit cycles produce real investment booms, and the later correction is concentrated in tangible capital.

5.7 Table 5: Predictable forecast errors drive cycles in debt issuance

	Year 1	Year 2	Year 3	Year 4	Year 5
<i>A. Dependent variable: Total debt issuance (change in long-term debt)</i>					
<i>macroPFE</i>	0.240*** (10.449)	0.134*** (5.990)	-0.020 (-0.946)	-0.123*** (-5.267)	-0.103*** (-3.976)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	56,282	51,506	46,763	41,809	37,425
R^2	.258	.216	.191	.177	.173
<i>B. Dependent variable: Change in long-term debt expiring in > 1 year</i>					
<i>macroPFE</i>	0.234*** (9.060)	0.127*** (4.982)	0.020 (0.702)	-0.067** (-2.579)	-0.112*** (-3.966)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	56,017	51,256	46,556	41,654	37,300
R^2	.224	.197	.182	.170	.168
<i>C. Dependent variable: Change in long-term debt expiring in < 1 year</i>					
<i>macroPFE</i>	0.380*** (7.844)	0.125*** (2.779)	-0.120*** (-2.816)	-0.215*** (-5.021)	-0.130*** (-2.917)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	56,910	52,042	47,270	42,301	37,872
R^2	.229	.144	.132	.127	.122

This table presents coefficient estimates from regressing total net debt issuance (panel A), change in long-term debt due to expire in more than one year (panel B) and change in long-term debt due to expire in less than one year (panel C) up to 5 years in the future, on aggregated predictable firm-level forecast error (*macroPFE*) and controls. Predictable firm-level forecast error (see Section 2 for details) is weighted by change in total debt (as in defined in panel A of Table 2). Column headings Year “k” (k = 1...5) mean the dependent variable is measured at time $t+k$, while all independent variables are measured at time t . Total net debt issuance is the change in long-term debt (“dltr”+“dlc” in Compustat) scaled by lagged total assets. Change in long-term debt due to expire in more than one year is “dltr” in Compustat scaled by lagged total assets. All specifications include firm fixed effects, firm-level controls (Tobin’s q, Cash flow to assets, log total assets, Cash to assets, Book leverage, Sales growth, ROA, default probability) and macro-level controls (the Leading Economic Index from the Conference Board, the Jurado, Ludvigson, and Ng [2015] index of macro uncertainty, the Baker and Wurgler [2006] sentiment index, the default spread, the risk-free rate, and the policy uncertainty index of Baker, Bloom, and Davis [2016]), as well as the current and lagged level of the dependent variable. Standard errors are bootstrapped according to [Abadie et al. \(2023\)](#). t -statistics are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Read:

- **Panel A.** A one-standard-deviation increase in *macroPFE* predicts +24.0% total long-term debt issuance in year $t + 1$ and +13.4% in year $t + 2$, followed by -12.3% and -10.3% in years $t + 4$ and $t + 5$.

Table 6
Predictable cycles in corporate investment: Conditioning on financial constraints

	<i>Dependent variable: Total investment</i>				
	Year 1	Year 2	Year 3	Year 4	Year 5
<i>A. Financial constraints measure is top quintile of Hadlock and Pierce (2010) index</i>					
<i>macroPFE</i>	0.050*** (5.583)	0.044*** (5.055)	0.011 (1.202)	-0.024** (-2.632)	-0.024*** (-3.042)
Financially constrained	-0.088 (-0.510)	0.040 (0.249)	-0.093 (-0.415)	0.336 (1.213)	-0.197 (-0.990)
<i>macroPFE</i> × Financially constrained	-0.008 (-0.262)	-0.0002 (0.027)	-0.010 (-0.237)	-0.029 (-0.823)	0.006 (0.160)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	57,597	52,648	47,150	42,139	37,643
R^2	.381	.315	.294	.289	.282
<i>B. Financial constraints measure is top quintile of Whited and Wu (2006) index</i>					
<i>macroPFE</i>	0.053*** (5.650)	0.046*** (4.992)	0.010 (0.959)	-0.037*** (-3.971)	-0.029*** (-3.214)
Financially constrained	0.167 (1.012)	-0.145 (-0.889)	-0.334* (-1.827)	-0.158 (-0.861)	-0.154 (-0.764)
<i>macroPFE</i> × Financially constrained	-0.035 (-1.365)	0.013 (0.463)	0.039 (1.504)	0.056* (1.821)	0.055 (1.631)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	52,458	48,015	42,964	38,336	34,169
R^2	.384	.321	.301	.297	.286
<i>C. Financial constraints measure is top quintile of Kaplan and Zingales (1997) index</i>					
<i>macroPFE</i>	0.049*** (5.614)	0.043*** (5.166)	0.007 (0.660)	-0.031*** (-3.614)	-0.019** (-2.184)
Financially constrained	-0.169 (-1.185)	-0.265* (-1.838)	-0.364** (-2.335)	-0.483*** (-2.822)	-0.075 (-0.541)
<i>macroPFE</i> × Financially constrained	-0.008 (-0.445)	0.005 (0.202)	0.022 (0.927)	0.026 (1.052)	-0.036 (-1.296)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	57,567	52,624	47,130	42,122	37,628
R^2	.381	.315	.294	.289	.281

This table presents coefficient estimates from regressing total investment up to 5 years in the future, on *macroPFE* (measured as in Section 2) and measures of financial constraints. Aggregated predictable firm-level forecast error (*macroPFE*) is calculated as predictable firm-level forecast error (see Section 2 for details), weighted by change in total debt (as in defined in panel A of Table 2). We also add interactions between all controls and a dummy variable that equals one for firms that are financially constrained. We categorize a firm as financially constrained if it falls in the top quintile for the financial constraints index of [Hadlock and Pierce \(2010\)](#) in panel A, the [Whited and Wu \(2006\)](#) measure in panel B, and the [Kaplan and Zingales \(1997\)](#) measure in panel C. Column headings Year “k” (k = 1...5) mean the dependent variable is measured at time $t+k$, while all independent variables are measured at time t . All specifications include firm fixed effects, firm-level controls (Tobin’s q, Cash flow to assets, log total assets, Cash to assets, Book leverage, Sales growth, ROA, default probability) and macro-level controls (the Leading Economic Index from the Conference Board, the Jurado, Ludvigson, and Ng [2015] index of macro uncertainty, the Baker and Wurgler [2006] sentiment index, the default spread, the risk-free rate, and the policy uncertainty index of Baker, Bloom, and Davis [2016]), as well as the current and lagged level of the dependent variable. Standard errors are bootstrapped according to [Abadie et al. \(2023\)](#). t -statistics are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

- **Panels B and C.** The boom-bust pattern holds both for debt maturing in more than one year and for the current portion of long-term debt.
- **Takeaway.** Financing cycles are even stronger than investment cycles, exactly as a Minsky-style credit mechanism would suggest.

5.8 Table 6: Investment cycles conditioning on financial constraints

Read:

- Across the Hadlock–Pierce, Whited–Wu, and Kaplan–Zingales measures, the interaction between macroPFE and financial constraints is generally tiny and statistically insignificant.
- The main boom-bust pattern in the coefficient on macroPFE itself survives almost unchanged.
- The table rejects the simple idea that only constrained firms expand investment when credit is cheap. Unconstrained firms do it too.

5.9 Table 7: Debt-issuance cycles conditioning on financial constraints

Table 7
Predictable cycles in debt issuance: Conditioning on financial constraints

	Dependent variable: Total debt issuance				
	Year 1	Year 2	Year 3	Year 4	Year 5
<i>A. Financial constraints measure is top quintile of Hadlock and Pierce (2010) index</i>					
macroPFE	0.242*** (10.137)	0.143*** (6.017)	−0.019 (−0.852)	−0.112*** (−4.546)	−0.090*** (−3.547)
Financially constrained	−0.375 (−0.859)	0.079 (0.244)	−0.145 (−0.318)	0.750 (1.164)	0.251 (0.402)
macroPFE × Financially constrained	0.017 (0.122)	−0.071 (−0.974)	0.008 (0.122)	−0.067 (−0.772)	−0.081 (−0.868)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	56,282	51,506	46,763	41,809	37,425
R ²	.259	.217	.191	.177	.173
<i>B. Financial constraints measure is top quintile of Whited and Wu (2006) index</i>					
macroPFE	0.253*** (9.860)	0.132*** (5.550)	−0.031 (−1.268)	−0.126*** (−4.658)	−0.094*** (−3.176)
Financially constrained	0.620 (1.487)	0.625 (1.371)	−0.752 (−1.625)	0.844 (1.610)	−0.444 (−0.738)
macroPFE × Financially constrained	−0.061 (−0.883)	−0.004 (−0.078)	0.091 (1.186)	−0.010 (−0.175)	0.021 (0.169)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	51,321	47,017	42,677	38,113	34,040
R ²	.264	.222	.200	.190	.178
<i>C. Financial constraints measure is top quintile of Kaplan and Zingales (1997) index</i>					
macroPFE	0.252*** (10.602)	0.151*** (6.219)	0.010 (0.353)	−0.097*** (−3.919)	−0.101*** (−3.771)
Financially constrained	−0.142 (−0.357)	−0.882** (−2.380)	−0.378 (−0.900)	−0.630 (−1.289)	−0.502 (−1.094)
macroPFE × Financially constrained	−0.075 (−1.294)	−0.114* (−1.787)	−0.183** (−2.620)	−0.156** (−2.229)	−0.023 (−0.317)
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	56,239	51,471	46,733	41,783	37,402
R ²	.261	.217	.192	.178	.173

This table presents coefficient estimates from regressing total net debt issuance up to 5 years in the future, on macroPFE (measured as in Section 2) and measures of financial constraints. Aggregated predictable firm-level forecast error (macroPFE) is calculated as predictable firm-level forecast error (see Section 2 for details), weighted by change in total debt (as in defined in panel A of Table 2). We also add interactions between all controls and a dummy variable that equals 1 for firms that are financially constrained. We categorize a firm as financially constrained if it falls in the top quintile for the financial constraints index of Hadlock and Pierce (2010) in panel A, the Whited and Wu (2006) measure in panel B, and the Kaplan and Zingales (1997) measure in panel C. Column headings Year “k” (k = 1, . . . , 5) mean the dependent variable is measured at time $t+k$, while all independent variables are measured at time t . Total net debt issuance is the change in long-term debt (“dltt”+“dlc” in Compustat) scaled by lagged total assets. All specifications include firm fixed effects, firm-level controls (Tobin’s q , Cash flow to assets, log total assets, Cash to assets, Book leverage, Sales growth, ROA, default probability) and macro-level controls (the Leading Economic Index from the Conference Board, the Jurado, Ludvigson, and Ng [2015] index of macro uncertainty, the Baker and Wurgler [2006] sentiment index, the default spread, the risk-free rate, and the policy uncertainty index of Baker, Bloom, and Davis [2016]), as well as the current and lagged level of the dependent variable. Standard errors are bootstrapped according to Abadie et al. (2023). t -statistics are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Read:

Table 8
Predictable firm-level forecast error drives cycles in physical investment: Conditioning on predictable firm-level forecast error tercile

	Year 1	Year 2	Year 3	Year 4	Year 5
<i>A. High versus low macroPFE</i>					
High tercile macroPFE	0.050*** (2.867)	0.032* (1.754)	−0.064*** (−3.67)	−0.072*** (−3.990)	−0.028 (−1.494)
Low tercile macroPFE	−0.028 (−1.535)	−0.028 (−1.596)	−0.010 (−0.500)	0.096*** (4.599)	0.170*** (8.158)
Observations	57,597	52,658	47,712	42,664	38,160
R ²	.397	.351	.346	.335	.337
<i>B. High versus low macroPFE conditioning on predictable firm-level forecast error tercile</i>					
High tercile macroPFE × high firmPFE	0.085*** (2.363)	0.126*** (3.471)	−0.044 (−1.260)	−0.043 (−1.362)	0.032 (0.777)
High tercile macroPFE × low firmPFE	0.053 (1.297)	0.062* (1.753)	−0.067* (−1.852)	−0.013 (−0.387)	0.039 (1.139)
Low tercile macroPFE × high firmPFE	−0.100*** (−2.488)	−0.060 (−1.444)	−0.016 (−0.310)	0.137*** (3.347)	0.199*** (4.650)
Low tercile macroPFE × low firmPFE	−0.046 (−1.377)	−0.048 (−1.200)	−0.010 (−0.192)	0.050 (1.201)	0.070* (1.813)
High macroPFE × high firmPFE − high macroPFE × low firmPFE	0.030 (0.629)	0.060 (1.427)	0.022 (0.480)	−0.027 (−0.601)	−0.010 (−0.254)
Low macroPFE × low firmPFE − low macroPFE × high firmPFE	−0.054 (−1.436)	−0.011 (−0.263)	−0.006 (−0.149)	0.089* (1.933)	0.113*** (2.872)
Observations	20,153	18,944	17,635	16,030	14,559
R ²	.432	.378	.368	.346	.343

This table presents coefficient estimates from regressing total physical investment up to 5 years in the future on a binary variables equal to one if aggregated predictable firm-level forecast error (macroPFE) is in the high or low tercile for the sample and controls. In panel B, we interact the binary tercile indicators for macroPFE with binary variables equal to one if the firm was in the highest or lowest tercile, respectively, for firmPFE in the year and zero otherwise. Predictable firm-level forecast error (see Section 2 for details) is weighted by change in total debt (as in defined in panel A of Table 2). Column headings Year “k” (k = 1, . . . , 5) mean the dependent variable is measured at time $t+k$, while all independent variables are measured at time t . The investment and Tobin’s q variables are measured as in Peters and Taylor (2017). Investment in physical capital is the change in physical capital divided by lagged total capital, where total capital is gross PPE (i.e., physical capital) plus the sum of goodwill, capitalized R&D, and capitalized SG&A (i.e., intangible capital). Tobin’s q is the market value of equity plus book value of debt divided by total capital. All specifications include firm fixed effects, firm-level controls (Tobin’s q , Cash flow to assets, log total assets, Cash to assets, Book leverage, Sales growth, ROA, default probability) and macro-level controls (the Leading Economic Index from the Conference Board, the Jurado, Ludvigson, and Ng [2015] index of macro uncertainty, the Baker and Wurgler [2006] sentiment index, the default spread, the risk-free rate, and the policy uncertainty index of Baker, Bloom, and Davis [2016]), as well as the current and lagged level of the dependent variable. Standard errors are bootstrapped according to Abadie et al. (2023). t -statistics are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

- The same message holds on the financing side: constrained and unconstrained firms show very similar debt-issuance cycles.
- The main coefficient on macroPFE remains strongly positive at short horizons and negative at longer horizons.
- This table reinforces the paper’s claim that both demand-side and supply-side optimism are needed, because a pure supply effect would have produced much stronger real responses among constrained firms.

5.10 Table 8: Physical-investment cycles conditioning on firm-level forecast-error terciles

Read:

- When aggregate optimism is high, firms in the highest firmPFE tercile increase physical investment more than firms in the lowest tercile: 8.5% versus 5.3% in year $t + 1$, and 12.6% versus 6.2% in year $t + 2$.
- When aggregate optimism is low, the mirror image holds: low-firmPFE firms cut investment more sharply first and then rebound more strongly later.
- The table is the cleanest cross-sectional mechanism test in the paper. It shows that the firms with the most extrapolative own expectations react the most over the cycle.

5.11 Table 9: Debt-issuance cycles conditioning on firm-level forecast-error terciles

Read:

- The symmetry from Table 8 carries over to financing. In optimistic states, high-firmPFE firms issue more debt than low-firmPFE firms in the boom years.
- In pessimistic states, low-firmPFE firms contract debt more sharply first and then show the stronger later rebound.
- This parallel between investment and financing is strong evidence that the same expectations mechanism is operating on both margins.

5.12 Figure 3: Impulse response functions from Q-theory

Read:

- The upper-left panel shows that diagnostic expectations create a much larger boom-bust response in I/K than rational expectations in the no-debt benchmark.
- The upper-right panel shows that adding riskless debt and financial frictions amplifies the response, but that model would also predict stronger heterogeneity by financial constraints than the data exhibit.
- The bottom panels show that the largest and most realistic fluctuations in both I/K and B/K occur when both managers and creditors have diagnostic expectations.

Table 9
Predictable firm-level forecast error drives cycles in total debt issuance: Conditioning on predictable firm-level forecast error tercile

	Year 1	Year 2	Year 3	Year 4	Year 5
<i>A. High versus low macroPFE</i>					
High tercile <i>macroPFE</i>	0.265*** (5.877)	0.250*** (5.380)	-0.105*** (-2.221)	-0.147*** (-3.019)	0.026 (0.431)
Low tercile <i>macroPFE</i>	-0.295*** (-6.479)	-0.251*** (-5.071)	-0.273*** (-4.884)	-0.036 (-0.620)	0.364*** (5.837)
Observations	56,282	51,506	46,763	41,809	37,425
R^2	.257	.216	.191	.176	.173
<i>B. High versus low macroPFE conditioning on predictable firm-level forecast error tercile</i>					
High tercile <i>macroPFE</i> × high tercile <i>firmPFE</i>	0.516*** (5.131)	0.527*** (5.112)	-0.116 (-1.195)	0.049 (0.485)	0.079 (0.736)
High tercile <i>macroPFE</i> × low tercile <i>firmPFE</i>	0.288*** (2.754)	0.282*** (2.622)	-0.172 (-1.577)	-0.048 (-0.389)	0.148 (1.180)
Low tercile <i>macroPFE</i> × low tercile <i>firmPFE</i>	-0.456*** (-4.413)	-0.439*** (-4.322)	-0.516*** (-4.396)	-0.043 (-0.427)	0.367*** (2.914)
Low tercile <i>macroPFE</i> × high tercile <i>firmPFE</i>	-0.331*** (-3.751)	-0.393*** (-4.069)	-0.431*** (-4.129)	-0.112 (-1.101)	0.276** (2.524)
high <i>macroPFE</i> × high <i>firmPFE</i> – high <i>macroPFE</i> × low <i>firmPFE</i>	0.226* (1.906)	0.240* (1.948)	0.062 (0.518)	0.096 (0.713)	-0.062 (-0.472)
low <i>macroPFE</i> × low <i>firmPFE</i> – low <i>macroPFE</i> × high <i>firmPFE</i>	-0.124 (-1.218)	-0.053 (-0.523)	-0.082 (-0.677)	0.064 (0.497)	0.086 (0.683)
Observations	19,838	18,651	17,369	15,811	14,373
R^2	.266	.226	.202	.189	.189

This table presents coefficient estimates from regressing total debt issuance up to 5 years in the future on a binary variables equal to one if aggregated predictable firm-level forecast error (*macroPFE*) is in the high or low tercile for the sample and controls. In panel B, we interact the binary tercile indicators for *macroPFE* with binary variables equal to one if the firm was in the highest or lowest tercile, respectively, for *firmPFE* in the year and zero otherwise. Predictable firm-level forecast error (see Section 2 for details) is weighted by change in total debt (as in defined in panel A of Table 2). Column headings Year “k” (k = 1...5) mean the dependent variable is measured at time $t+k$, while all independent variables are measured at time t . Total net debt issuance is the change in long-term debt (“dltt”+“dlc” in Compustat) scaled by lagged total assets. Tobin’s q is measured as in Peters and Taylor (2017), the market value of equity plus book value of debt divided by total capital. All specifications include firm fixed effects, firm-level controls (Tobin’s q, Cash flow to assets, log total assets, Cash to assets, Book leverage, Sales growth, ROA, default probability) and macro-level controls (the Leading Economic Index from the Conference Board, the Jurado, Ludvigson, and Ng [2015] index of macro uncertainty, the Baker and Wurgler [2006] sentiment index, the default spread, the risk-free rate, and the policy uncertainty index of Baker, Bloom, and Davis [2016]), as well as the current and lagged level of the dependent variable. Standard errors are bootstrapped according to Abadie et al. (2023). t -statistics are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01, 0.01$.

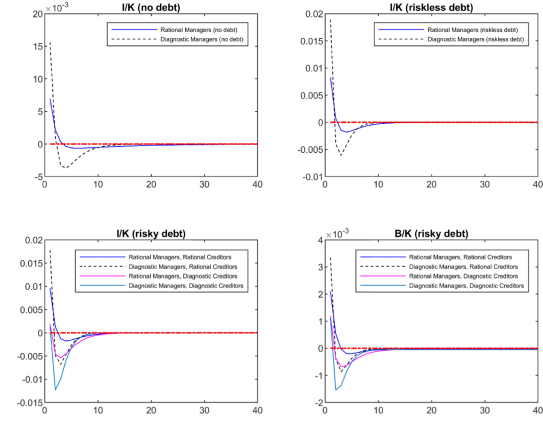


Figure 3
Impulse response functions
This figure plots the impulse response functions (IRFs) of neoclassical Q-theory under rational expectations (RE) and diagnostic expectations (DE). All variables are measured at time $t=0$. Year “k” (k = 1...5) means the dependent variable is measured at time $t+k$. All IRFs are plotted as their steady-state values. Hence, the economic magnitude (y-axis) can be interpreted as the change (expressed as % of the mean) in the dependent variable caused by a one-standard-deviation shock to TFP. The x-axis represent the steady-state value of the variable.

- The figure summarizes the paper’s theoretical punchline: biased beliefs, not only financial frictions, are needed to match the data.

6 Short summary

- **Conclusion.** Predictable overextrapolation in long-term earnings expectations helps explain both aggregate credit booms and firm-level boom-bust cycles in borrowing and investment.
- **Intuition.** When firms and credit investors become too optimistic after good news, credit becomes too cheap, firms issue more debt and invest more, and later disappointment forces reversal.
- **Empirical lesson.** Measuring biased expectations directly — rather than inferring sentiment only from prices or spreads — yields a powerful way to study the real effects of credit cycles.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The paper shows that a debt-weighted measure of predictable analyst forecast errors captures the irrational-expectations component of credit cycles.
- **Magnitude.** The time-series relation is large: the index correlates 0.75 with credit market sentiment and predicts sizable boom-bust patterns in both investment and debt issuance.
- **Credibility.** The evidence is supported by validation against excess bond premia and Blue Chip credit-yield forecast errors, by rich macro and firm controls, and by cross-sectional patterns

that align closely with the theory.

- **Mechanism evidence.** Similar responses across constrained and unconstrained firms imply that both investors and managers are behaving non-rationally during credit booms.
- **Cross-sectional discipline.** Firms with the most optimistic own predictable forecast errors react the most in optimistic aggregate states, and the mirror image holds in pessimistic states.
- **Theoretical implication.** Standard rational-expectations financial-frictions models are not enough. A dynamic model with diagnostic expectations by both managers and creditors fits the joint facts on credit spreads, issuance, and investment much better.

7.2 Economic intuition

- The paper's mechanism starts with overextrapolation: good news leads analysts, managers, and creditors to expect too much persistence in future earnings and credit quality.
- Because the optimistic firms are also the ones issuing debt, the biased-expectations shock directly enters the credit market. Cheap credit raises debt issuance first and investment next.
- The reversal is endogenous. When the extrapolated good news fails to materialize, both credit conditions and firm policies correct, producing predictable declines in borrowing and real investment several years later.
- The absence of stronger responses among constrained firms is informative: it means the boom is not just a relaxation of borrowing constraints for the most constrained firms, but a broader optimism shock that affects financing demand as well as financing supply.

7.3 Takeaway

This paper gives a concrete empirical implementation of the Minsky idea that credit cycles are belief cycles. Its contribution is not only to show that sentiment predicts bad times later, but to trace the full chain from biased analyst expectations to credit-market sentiment, debt issuance, and corporate investment. The overall lesson is that belief formation belongs at the center of any realistic theory of credit-driven business fluctuations.